Lucas Coefficient Families and Exact Arithmetic for Every Integer Coefficient

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Abstract

We study weighted Lucas expressions for positive integer coefficients. Every positive difference $L_i - 5F_{2r}L_j$, for $r \geq 1$, has a unique representation; we determine its admissible indices, bounded-index count, and asymptotics. In contrast, for every $q = L_{2r}$, both $qL_i - L_j$ and $qL_i + L_j$ have infinitely many multiply represented positive values. A transfer lemma relates sum collisions to reversed-difference collisions. An algebraic-norm bound gives exact representation and rank algorithms for every integer coefficient, and excludes all positive simultaneous-shift collision families for coefficients at least two.

Let $F_0 = 0, F_1 = 1$ and $L_0 = 2, L_1 = 1$, with $u_{n+2} = u_{n+1} + u_n$. Use $F_{-n} = (-1)^{n+1}F_n$ and $L_{-n} = (-1)^n L_n$. Put

$$S_q = \{L_i - qL_j > 0 : i, j \geq 1\}, \quad T_q = \{qL_i - L_j > 0 : i, j \geq 1\}, \quad P_q = \{qL_i + L_j : i, j \geq 1\}.$$

The recurrence gives

$$L_{m+4} - L_m = 5F_{m+2}, \qquad F_a L_b = F_{a+b} + (-1)^b F_{a-b}. \tag{1}$$

The collision-free coefficients begin $5, 15, 40, 105, 275, \ldots$; the persistent-collision coefficients begin $3, 7, 18, 47, 123, \ldots$.

An infinite collision-free family

Theorem 1. *Fix $r \geq 1$ and set $q = 5F_{2r}$. Then*

$$L_i - qL_j > 0 \quad \Longleftrightarrow \quad (i, j) = (2r + 2, 1) \quad \text{or} \quad i \geq 2r + 3,\ 1 \leq j \leq i - 2r - 2.$$

Every positive integer in S_q has precisely one representation.

Proof. The recurrence yields

$$q = 5F_{2r} = L_{2r+2} - L_{2r-2}, \quad q - L_{2r+1} = L_{2r-1} > 0, \quad L_{2r+2} - q = L_{2r-2} > 0. \tag{2}$$

Thus $j = 1$ first becomes admissible at $i = 2r + 2$. For $j \geq 2$ put $A_j = L_{j+2r+2} - qL_j$ and $B_j = qL_j - L_{j+2r+1}$. Both obey the Lucas recurrence in j. Checking two initial values in r and using the recurrence in steps of two gives

$$A_2 = L_{2r+4} - 3q = L_{2r-4} > 0, \quad A_3 = A_2 + A_1 > 0, \quad A_1 = L_{2r+3} - q > 0,$$

with $L_{-2} = 3$ when $r = 1$. For B_j, the first two terms of the sequences $B_2(r)$ and $B_3(r)$ are

$$(B_2(1), B_2(2)) = (4, 16), \qquad (B_3(1), B_3(2)) = (2, 13).$$

Each satisfies $b_{r+2} = 3b_{r+1} - b_r$ as a function of r. An increasing positive initial pair remains increasing and positive. Therefore $B_2, B_3 > 0$ for all $r \geq 1$; the recurrence in j then gives $A_j, B_j > 0$ for all $j \geq 2$. Hence $L_{j+2r+1} < qL_j < L_{j+2r+2}$ for $j \geq 2$, establishing the stated index region.

Let $\mathcal{B}_i$ be the positive values with first index i. They are distinct within each block, and $\max \mathcal{B}_i = L_i - q$ for $i \geq 2r + 2$. Using (1) with $b = i - 2r + 2$, the next block of the same residue class satisfies

$$\begin{aligned} \min \mathcal{B}_{i+4} - \max \mathcal{B}_i &= L_{i+4} - L_i - qL_{i-2r+2} + q \\ &= q + 5F_{i-4r+2} > 0. \end{aligned} \tag{3}$$

For a nonnegative Fibonacci index the inequality is immediate. For a negative index, $i \geq 2r + 2$ implies $0 < |i - 4r + 2| \leq 2r - 4 < 2r$, so $|F_{i-4r+2}| < F_{2r}$. Moreover $A_{j+4} > A_j$ follows from the strictly positive Lucas recurrence; thus blocks separated by any positive multiple of four are strictly ordered.

Finally, $(L_1, L_2, L_3, L_4) \bmod 5 = (1, 3, 4, 2)$ is the repeating residue cycle. Since $5 \mid q$, two equal differences must have first indices congruent modulo four. They cannot belong to different such blocks, by (3), and the same block has distinct values. □

Corollary 2. *For $N \geq 2r + 2$ the number of distinct positive differences with first index at most N equals*

$$1 + \frac{(N - 2r - 2)(N - 2r - 1)}{2}.$$

Writing $\varphi = (1 + \sqrt{5})/2$ and $A_r(x) = \#(S_{5F_{2r}} \cap [1, x])$,

$$A_r(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O_r(\log x), \qquad \log_\varphi b_{r,n} = \sqrt{2n} + O_r(1),$$

where $b_{r,n}$ is the n-th positive value. Its exact value can be recovered by integer binary search.

Proof. Summing block sizes $1, 1, 2, 3, \ldots$ gives the formula. Let $m(x) = \max\{m \geq 1 : L_m \leq x\} = \log_\varphi x + O(1)$. All positive differences with $i \leq m(x)$ are at most x. For $j \geq 2$, $A_2 \geq L_0 = 2$ and $A_3 \geq L_1 = 1$, so recurrence gives $A_j \geq L_{j-2}$. Therefore, for all sufficiently large i, a value in $\mathcal{B}_i$ is at least L_{i-2r-4}; if it is at most x then $i \leq m(x) + 2r + 4$. Applying the triangular count to these two index bounds proves the asymptotic. Inverting it proves the nth-term estimate. An exact count tests $L_j \geq \lceil (L_i - x)/q \rceil$ within finitely many admissible blocks; binary search inverts that monotone count. □

A family with infinitely many collisions

Theorem 3. *For every $r \geq 1$, the coefficient $q = L_{2r}$ has infinitely many repeated positive values in both T_q and P_q. For every $n \geq 1$,*

$$L_{n+4r} = qL_{n+2r} - L_n = qL_{n+6r} - L_{n+8r} > 0, \tag{4}$$

$$qL_{n+2r} + L_{n+8r} = qL_{n+6r} + L_n. \tag{5}$$

The two index pairs in each equality are distinct.

Proof. The product identity $L_aL_b = L_{a+b} + (-1)^a L_{b-a}$ with $a = 2r$ gives

$$qL_{n+2r} = L_{n+4r} + L_n, \quad qL_{n+6r} = L_{n+8r} + L_{n+4r}.$$

These yield (4) and (5). Both resulting sequences of positive values increase strictly with n, providing infinitely many collisions. □

Proposition 4 (Collision transfer)**.** *Suppose S_q has unique positive representations and q divides no Lucas number L_j with $j \geq 1$. Then every sum collision $qL_a + L_b = qL_c + L_d$ with $a < c$ induces a positive collision $qL_a - L_d = qL_c - L_b > 0$ in T_q. These hypotheses hold for every $q = 5F_{2r}$.*

Proof. Set $t = qL_a - L_d = qL_c - L_b$. If $t = 0$, then $q \mid L_d$, a contradiction. If $t < 0$, the equality $L_d - qL_a = L_b - qL_c > 0$ gives two representations in S_q, also a contradiction. Thus $t > 0$. The final assertion follows from Theorem 1 and the four nonzero Lucas residues modulo five. □

An exact algorithm for every integer coefficient

The coefficient families above admit efficient special-purpose rankings. An integer-bound argument gives complete, exact arithmetic for every coefficient, with no conjectural collision classification needed.

Theorem 5 (Uniform finite search bounds)**.** *Let $q \geq 1$ and $X \geq 1$ be integers, and put*

$$B = (q+1)(X+q+1) + 1.$$

For any $0 < L_i - qL_j \leq X$ with $i, j \geq 1$, necessarily

$$L_j \leq B, \qquad L_i \leq X + qB. \tag{6}$$

For $q \geq 2$ and $0 < qL_i - L_j \leq X$, necessarily

$$L_i \leq M := \max\{B, \lfloor X/(q-1) \rfloor\}, \qquad L_j \leq qM. \tag{7}$$

When $q = 1$, the expressions $qL_i - L_j$ and $L_i - qL_j$ coincide and obey (6). For $q \geq 1$ and $0 < qL_i + L_j \leq X$,

$$L_i \leq \lfloor (X-1)/q \rfloor, \qquad L_j \leq X - q. \tag{8}$$

Consequently, for every *$q \in \mathbb{Z}$, the set of distinct positive values at most X in each of S_q, T_q, P_q is effectively computable by finite integer arithmetic. For every nonzero q, all representations of each positive value can also be enumerated, yielding its exact multiplicity. For $q = 0$, the positive sets S_0 and P_0 are the positive-index Lucas numbers, each with infinitely many representations; T_0 is empty.*

Proof. Write $\alpha = \varphi$ and $\beta = (1 - \sqrt{5})/2 = -1/\varphi$. For $d \geq 1$ and $q \geq 1$, the algebraic integer $\alpha^d - q$ has nonzero integral norm

$$(\alpha^d - q)(\beta^d - q) \in \mathbb{Z} \setminus \{0\}.$$

It cannot vanish because $\alpha^d = F_d\alpha + F_{d-1}$ is irrational. Since $|\beta^d - q| \leq q + 1$, this gives

$$|\alpha^d - q| \geq \frac{1}{q+1}. \tag{9}$$

A positive difference $L_i - qL_j$ has $i > j$; write $d = i - j$. By Binet's formula,

$$L_i - qL_j = \alpha^j(\alpha^d - q) + \beta^i - q\beta^j.$$

For $0 < L_i - qL_j \le X$, it follows that

$$\frac{\alpha^j}{q+1} \le X + |\beta^i - q\beta^j| \le X + q + 1.$$

Thus $\alpha^j \le (q+1)(X+q+1)$, and $L_j = \alpha^j + \beta^j \le B$. The equality $L_i = x + qL_j$ now proves (6).

For $0 < qL_i - L_j \le X$, when $j > i$, the same argument with $d = j - i$ shows $L_i \le B$, hence $L_j = qL_i - x \le qB$. When $j \le i$ and $q \ge 2$, $x = qL_i - L_j \ge (q-1)L_i$; this proves (7). The sum bounds (8) follow from $L_i, L_j \ge 1$.

For negative coefficients, write $m = -q > 0$. Then $S_{-m} = P_m$ with the indices interchanged, $P_{-m} = S_m$ with the indices interchanged, and T_{-m} is empty. The assertions for $q = 0$ are immediate. Thus the finite index bounds cover all integers. □

Corollary 6 (Universal exact counting and unranking)**.** *For any integer q, expression type S, T or P, and integer $X \ge 0$, there is a terminating algorithm for the number of* distinct *positive values at most X, exact membership, and all positive representations (or detection of infinite multiplicity when $q = 0$). Whenever the positive set is nonempty, its n-th distinct value is computable exactly for every $n \ge 1$.*

Proof. Enumerate the finitely many indices specified by Theorem 5, retain the positive values at most X, and deduplicate integer values. Each nonempty positive set is unbounded: for S_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, for T_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, and for P_q with $q > 0$ fix $i = 1$ and let $j \to \infty$. The negative and zero cases reduce to these or the Lucas numbers themselves. Therefore doubling the threshold until n values are found and then binary-searching the exact count always terminates. The only empty case is T_q for $q \le 0$. □

Proposition 7 (No simultaneous-shift positive collisions)**.** *For every integer $q \ge 2$, there are no indices $i < k$, $j < \ell$ such that*

$$L_{n+i} - qL_{n+j} = L_{n+k} - qL_{n+\ell} > 0 \quad \textit{for every } n \ge 0.$$

Proof. An equality for all shifts implies, by Binet's formula, both its α- and β-coefficient equalities. Put $h = k - i \ge 1$, $e = \ell - j \ge 1$, and $s = i - j$. Positivity for large n forces $\alpha^s > q$, so $s \ge 2$, and

$$\alpha^s(\alpha^h - 1) = q(\alpha^e - 1)$$

forces $e > h$, hence $e \ge 2$. Conjugation gives

$$q|\beta^e - 1| = |\beta|^s|\beta^h - 1| \le \alpha^{-2}(1 + \alpha^{-1}) = \alpha^{-1}.$$

But $q|\beta^e - 1| \ge 2(1 - \alpha^{-2}) = 2/\alpha > \alpha^{-1}$, a contradiction. □

Context. The case $q = 5F_2 = 5$ generalizes the unique representation of positive differences in OEIS A191981. The two coefficient families provide explicit sufficient conditions for uniqueness and for infinite collisions. The classification of all integer coefficients remains open.